The decomposition (following Balitsky–Chirilli, eqs. (59), (62), (63)) is

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \underbrace{\Big(\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} - \frac{d^3N^{\mathrm{sub}}_{2\,\mathrm{NLO}}}{d^3k}\Big)}_{\text{(I) finite}} + \underbrace{\frac{d^3N^{\mathrm{sub,\,UV}}_{2\,\mathrm{NLO}}}{d^3k}}_{\text{(II) UV}} + \underbrace{\frac{d^3N^{\mathrm{sub,\,fin}}_{2\,\mathrm{NLO}}}{d^3k}}_{\text{(III) finite}}. \tag{1}$$

## (I) $N_2 - N_2^{\mathrm{sub}}$: UV finite (colour vanishes at $z = x$), $d_\perp = 2$

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} - \frac{d^3N^{\mathrm{sub}}_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)} \\
&\times e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\big[(y-x)^2-(y-z)^2\big]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} \\
&-\frac{p^+}{(k^+-p^+)}\,\Bigg[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\Bigg] \\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\big[(y-x)^2-(y-z)^2\big]}{2\,(x-z)^2}\Bigg] \\
&-\frac{(k^+-p^+)}{p^+}\,\Bigg[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\Bigg] \\
&+\frac{p^+\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\big[(y-x)^2-(y-z)^2\big]}{2\,(x-z)^2}\Bigg] \\
&-\Bigg[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\Bigg]\Bigg] \\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^f(y)
\end{aligned} \tag{2}$$

## (II) UV part of $N_2^{\mathrm{sub}}$ (dim. reg., $d_\perp = 2-\epsilon$, $\overline{MS}$)

$$\begin{aligned}
\frac{d^3N^{\mathrm{sub,\,UV}}_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\Big[2\log\Big(\frac{k^+-\Lambda}{\Lambda}\Big)-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}\Big]\int_{x,y,x',w'} e^{-ik\cdot(w'-x)} \\
&\times\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-x)^2\bar\mu^2}{4}\Big]\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned} \tag{3}$$

## (III) Finite remainder of $N_2^{\mathrm{sub}}$, $d_\perp = 2$

$$\begin{aligned}
\frac{d^3N^{\mathrm{sub,\,fin}}_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}\Bigg\{ \\
&e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\big[(y-x)^2-(y-z)^2\big]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} \\
&-\frac{p^+}{(k^+-p^+)}\,\Bigg[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\Bigg] \\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\big[(y-x)^2-(y-z)^2\big]}{2\,(x-z)^2}\Bigg] \\
&-\frac{(k^+-p^+)}{p^+}\,\Bigg[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\Bigg] \\
&+\frac{p^+\,\delta_{im}}{k^+}\,\Bigg[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\big[(y-x)^2-(y-z)^2\big]}{2\,(x-z)^2}\Bigg]
\end{aligned}$$

$$
- \left[ \frac{(y-x)^m\,(x-z)^i}{(x-z)^2} - \frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2} + \frac{(y-z)^m\,(x-z)^i}{(x-z)^2} + \frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2} \right] \Bigg]
$$

$$
- e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[ \frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+} \right] \frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2} \Bigg\}
$$

$$
\times \Big[ U^{ab'}(x') - U^{ab'}(w') \Big]\, N_c \Big[ U^{af}(y) - U^{af}(x) \Big] \rho^{b'}(x')\,\rho^f(y) \tag{4}
$$

## Notes

- (II) is the $d$-dimensional $z$ integral of the $z \to x$ limit of the subtraction kernel,

$$
\frac{1}{k^+} P(\xi)\, e^{-ik\cdot(w'-x)} \frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2} \frac{(y-z)\cdot(y-x)}{(y-z)^2(x-z)^2},
$$

  with $\mu^{-\epsilon}\int d^{2-\epsilon}z\,\frac{(y-z)\cdot(y-x)}{(y-z)^2(z-x)^2} = \pi\big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-x)^2\bar{\mu}^2}{4}\big]$, $\bar{\mu}^2 = 4\pi e^{-\gamma_E}\mu^2$, and $\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda} dp^+ P = 2\log\frac{k^+-\Lambda}{\Lambda} - \frac{11}{6} + O(\Lambda)$.

- (III) is finite: the last line cancels the $1/(x-z)^2$ of the lines above it. They must stay together under one $\int_z$; integrating the last line alone over $z$ in dim. reg. would just give $-$(II).

- (I)+(II)+(III) $= N_2$ exactly. The only $\log\mu$ is in (II).